

Week 4 – §4.1 & 4.2: Afhankelijkheid, rang, dimensie en basis

Studiegids uit de HC-aantekeningen (dictaat-nummers ter referentie)

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*Lecture framing: last week we had $\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_r)$. Today’s question: **how do we find, by a method instead of “staren” (staring), the minimal number of vectors needed to span a subspace W ?** That number is the **dimension**; such a minimal set is a **basis**.*

0. The motivating example (VB in $\mathbb{R}^4$)

$$W = \text{Span} \left(\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \\ 2 \\ 0 \end{pmatrix} \right).$$

By “staring” we see

$$\begin{pmatrix} 3 \\ 3 \\ 2 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 2 \\ 0 \end{pmatrix}, \quad \text{equivalently} \quad 2 \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 2 \\ 0 \end{pmatrix} - \begin{pmatrix} 3 \\ 3 \\ 2 \\ 0 \end{pmatrix} = \mathbf{0}.$$

Vocabulary this gives us: - the third vector is a **lineaire combinatie** of the first two; the three vectors are **afhankelijk**; - the second equation is a **niet-triviale relatie** between the three vectors.

Hence $W = \text{Span}((1, 1, 0, 0)^t, (1, 1, 2, 0)^t)$; those two are **onafhankelijk** — they form a **basis** of W — and with fewer than two vectors W cannot be spanned: $\dim W = 2$.

Topics of the week, in order: 1. lineaire combinatie, 2. lineair (on)afhankelijk, 3. lineaire relatie (triviaal / niet-triviaal) + *how to find relations*, 4. rang / dimensie, 5. basis, then the theorems of §4.2.

1. Definities (given $\mathbf{v}_1, \dots, \mathbf{v}_r \in \mathbb{R}^n$)

#	Dictaat	Definition
1	§3.4 (3.4.3)	$\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_r) = \{\lambda_1 \mathbf{v}_1 + \dots + \lambda_r \mathbf{v}_r \mid \lambda_i \in \mathbb{R}\}$ is the opspansel of $\mathbf{v}_1, \dots, \mathbf{v}_r$.
2	§4.1 (4.1.1)	$\lambda_1 \mathbf{v}_1 + \dots + \lambda_r \mathbf{v}_r$ ($\lambda_i \in \mathbb{R}$) is a lineaire combinatie of $\mathbf{v}_1, \dots, \mathbf{v}_r$. (These are exactly the elements of the span.)
3	§4.1 (4.1.1)	$\mathbf{w} \in \mathbb{R}^n$ is lineair afhankelijk van $\mathbf{v}_1, \dots, \mathbf{v}_r$ if there are λ_i with $\mathbf{w} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_r \mathbf{v}_r$. Equivalently: $\mathbf{w} \in \text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_r)$; equivalently: $\mathbf{w}$ can be written as a linear combination of them.
4	§4.1 (4.1.1)	The r -tuple $\mathbf{v}_1, \dots, \mathbf{v}_r$ is lineair afhankelijk if at least one $\mathbf{v}_i$ is linearly dependent on the other $r - 1$ vectors.
5	§4.1 (4.1.1)	$\mathbf{v}_1, \dots, \mathbf{v}_r$ are onafhankelijk if they are not afhankelijk.
6	§4.1 (4.1.2)	An equation $\lambda_1 \mathbf{v}_1 + \dots + \lambda_r \mathbf{v}_r = \mathbf{0}$ is a lineaire relatie ($\lambda_1, \dots, \lambda_r$). It is triviaal if all $\lambda_i = 0$; niet-triviaal if at least one $\lambda_i \neq 0$.

Examples from the lecture: - $(3, 3, 2, 0)^t$ is linearly dependent on $(1, 1, 0, 0)^t$ and $(1, 1, 2, 0)^t$ (def. 3).
- $\{(1, 1, 0, 0)^t, (1, 1, 2, 0)^t, (3, 3, 2, 0)^t\}$ is an afhankelijk stel (def. 4). - $\{(1, 1, 0, 0)^t, (1, 1, 2, 0)^t\}$ is an onafhankelijk stel: neither is a multiple of the other (def. 5).

The trivial relation always exists ($\lambda_i = 0$ for all i), so it carries no information. The interesting question is always: **is there a non-trivial relation?**

Slide quiz. Find $\lambda_1, \lambda_2, \lambda_3$ with $\lambda_1 \mathbf{e}_1 + \lambda_2 \mathbf{e}_2 + \lambda_3 \mathbf{e}_3 = \mathbf{0}$ in $\mathbb{R}^3$. “Met de hand”: looking at each coordinate gives $\lambda_1 = \lambda_2 = \lambda_3 = 0$ — only the trivial relation, so $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ are independent. (The general method is below: use H3.)

2. Afhankelijk $\iff$ niet-triviale relatie (Stelling 4.1.3 / 4.1.5)

Conclusie 1 ($\Rightarrow$). Suppose $\mathbf{v}_1, \dots, \mathbf{v}_r$ is dependent, say $\mathbf{v}_1 = \lambda_2 \mathbf{v}_2 + \dots + \lambda_r \mathbf{v}_r$. Bring everything to one side:

$$1 \cdot \mathbf{v}_1 - \lambda_2 \mathbf{v}_2 - \dots - \lambda_r \mathbf{v}_r = \mathbf{0},$$

a relation with $\lambda_1 = 1 \neq 0$: **non-trivial**. So: *afhankelijk $\Rightarrow$ there exists a non-trivial relation*.

($\Leftarrow$). Suppose $\lambda_1 \mathbf{v}_1 + \dots + \lambda_r \mathbf{v}_r = \mathbf{0}$ with not all $\lambda_i = 0$; say $\lambda_1 \neq 0$. Then $\lambda_1 \mathbf{v}_1 = -\lambda_2 \mathbf{v}_2 - \dots - \lambda_r \mathbf{v}_r$, and because $\lambda_1 \neq 0$ we may divide:

$$\mathbf{v}_1 = -\frac{\lambda_2}{\lambda_1} \mathbf{v}_2 - \dots - \frac{\lambda_r}{\lambda_1} \mathbf{v}_r,$$

so $\mathbf{v}_1$ depends on $\mathbf{v}_2, \dots, \mathbf{v}_r$: the set is dependent.

Conclusie 2 (Stelling 4.1.3). $\mathbf{v}_1, \dots, \mathbf{v}_r$ afhankelijk $\iff$ there exists a **niet-triviale** relation between them.

Conclusie 3 (Stelling 4.1.5). $\mathbf{v}_1, \dots, \mathbf{v}_r$ onafhankelijk $\iff$ the **only** relation between them is the trivial one.

Conclusie 3 is the test you use in practice: to prove independence, show that $\lambda_1 \mathbf{v}_1 + \dots + \lambda_r \mathbf{v}_r = \mathbf{0}$ forces all $\lambda_i = 0$.

3. How do we *find* non-trivial relations? $\rightarrow$ Gauss–Jordan on $A\lambda = \mathbf{0}$

VB. $\{(1, 1, 0, 0)^t, (1, 1, 2, 0)^t, (3, 3, 2, 0)^t\}$. We look for $\lambda_1, \lambda_2, \lambda_3$ with

$$\lambda_1 \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 1 \\ 2 \\ 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} 3 \\ 3 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

(If only $\lambda_1 = \lambda_2 = \lambda_3 = 0$ works: independent. If some $\lambda_i \neq 0$ works: we have found a dependence.)

Rewriting coordinate by coordinate gives a **system of 4 equations in 3 unknowns**:

$$\begin{array}{lcl} 1\lambda_1 + 1\lambda_2 + 3\lambda_3 = 0 \\ \lambda_1 + \lambda_2 + 3\lambda_3 = 0 \\ 0\lambda_1 + 2\lambda_2 + 2\lambda_3 = 0 \\ 0\lambda_1 + 0\lambda_2 + 0\lambda_3 = 0 \end{array} \quad \text{i.e.} \quad \underbrace{\begin{pmatrix} 1 & 1 & 3 \\ 1 & 1 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{pmatrix}}_{\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \text{ as columns}} \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

In het algemeen. All relations between $\mathbf{v}_1, \dots, \mathbf{v}_r$ are found by solving

$$\boxed{A\lambda = \mathbf{0}}, \quad A = \text{the matrix with } \mathbf{v}_1, \dots, \mathbf{v}_r \text{ as columns.}$$

This is a **homogeneous system**; its solution set S_{hom} is also called the **nulruimte** of A : $\text{Nul}(A) = \{\mathbf{x} \in \mathbb{R}^r \mid A\mathbf{x} = \mathbf{0}\}$. Solve it with Gauss–Jordan:

$$(A \mid \mathbf{0}) \sim \dots \sim (A' \mid \mathbf{0}).$$

The last column stays $\mathbf{0}$ throughout, so you may leave it out. And since row operations do not change S_{hom} :

$$\boxed{\text{relaties tussen de kolommen van } A \xleftrightarrow{1-1} \text{relaties tussen de kolommen van } A'}$$

In A' (volledig rijgereduceerd) the relations can be **read off directly**.

Worked example (from the board)

$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ with $\mathbf{v}_1 = (1, 1, -2)^t$, $\mathbf{v}_2 = (2, 1, -3)^t$, $\mathbf{v}_3 = (7, 5, -12)^t$, $\mathbf{v}_4 = (11, 5, -16)^t$. What are the relations?

$$\begin{pmatrix} 1 & 2 & 7 & 11 \\ 1 & 1 & 5 & 5 \\ -2 & -3 & -12 & -16 \end{pmatrix} \xrightarrow[r_3 \rightarrow r_3 + 2r_1]{r_2 \rightarrow r_2 - r_1} \begin{pmatrix} 1 & 2 & 7 & 11 \\ 0 & -1 & -2 & -6 \\ 0 & 1 & 2 & 6 \end{pmatrix}$$

$$\begin{array}{c} r_3 \rightarrow r_3 + r_2 \\ r_2 \rightarrow -r_2 \end{array} \begin{pmatrix} 1 & 2 & 7 & 11 \\ 0 & 1 & 2 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{r_1 \rightarrow r_1 - 2r_2} \begin{pmatrix} \boxed{1} & 0 & 3 & -1 \\ 0 & \boxed{1} & 2 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Read off from A' : - column 3 = 3·column 1 + 2·column 2: $(3, 2, 0)^t = 3(1, 0, 0)^t + 2(0, 1, 0)^t$, so **also** $\mathbf{v}_3 = 3\mathbf{v}_1 + 2\mathbf{v}_2$, i.e. $(7, 5, -12)^t = 3(1, 1, -2)^t + 2(2, 1, -3)^t$. ✓ - column 4 = -1·column 1 + 6·column 2: $(-1, 6, 0)^t = -(1, 0, 0)^t + 6(0, 1, 0)^t$, so **also** $\mathbf{v}_4 = -\mathbf{v}_1 + 6\mathbf{v}_2$, i.e. $(11, 5, -16)^t = -(1, 1, -2)^t + 6(2, 1, -3)^t$. ✓

(Why “also”? Because a relation $A\lambda = \mathbf{0}$ is a solution of the homogeneous system, and the solution set is preserved by Gauss–Jordan.)

Conclusion. $\mathbf{v}_1, \mathbf{v}_2$ are independent (the pivot columns: no relation with only λ_1, λ_2 non-zero) and they suffice to build every vector of S as a linear combination, since $\mathbf{v}_3, \mathbf{v}_4$ depend on them. So $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis of $\text{Span}(S)$ and the rank is 2.

***Recipe (= Stelling 4.2.8).** Put the vectors as **columns** of A , row-reduce. The vectors in the **pivot columns** form a basis of the span; the number of pivots is the rank. Each non-pivot column of A' tells you how the corresponding original vector is built from the pivot vectors. **Let op:** Take the basis vectors from the **original** A , not from A' — the rows have been mixed, so the columns of A' are different vectors.*

4. Rang, dimensie, basis (§4.2)

Def. 7 (4.2.1). Let $S \subseteq \mathbb{R}^n$, $S \neq \emptyset$. The **largest** number r such that S contains an r -tuple of **independent** vectors is the **rang** of S : $\text{rang}(S) = r$. If S is a lineaire deelruimte we also call the rank the **dimensie**: $\dim(S)$. If $S = \{\mathbf{0}\}$ then $\dim(S) = 0$.

Def. 8 (4.2.3). Let $S \subseteq \mathbb{R}^n$. A finite subset $B \subseteq S$ is a **basis** of S if 1. B consists of **independent** vectors, and 2. every $\mathbf{v} \in S$ can be written as a linear combination of elements of B , i.e. $S \subseteq \text{Span}(B)$.

In the motivating example: $B = \{(1, 1, 0, 0)^t, (1, 1, 2, 0)^t\}$ is a basis of W , $\dim W = 2$. For $\mathbb{R}^n$ itself, $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a basis (the standaardbasis) and $\dim \mathbb{R}^n = n$.

5. The theorems of §4.1–4.2, as explained in the lecture

Stelling 4.1.7 — coordinates w.r.t. a basis are unique

If $\mathbf{b}_1, \dots, \mathbf{b}_r$ are independent and $\mathbf{v}$ depends on them, then $\mathbf{v}$ can be written as a linear combination of $\mathbf{b}_1, \dots, \mathbf{b}_r$ in exactly one way.

Bewijs. Suppose $\mathbf{v} = \lambda_1 \mathbf{b}_1 + \dots + \lambda_r \mathbf{b}_r$ and also $\mathbf{v} = \mu_1 \mathbf{b}_1 + \dots + \mu_r \mathbf{b}_r$. Subtracting:

$$(\lambda_1 - \mu_1)\mathbf{b}_1 + \dots + (\lambda_r - \mu_r)\mathbf{b}_r = \mathbf{0}.$$

This is a relation between independent vectors, so it must be the trivial one: $\lambda_i - \mu_i = 0$, i.e. $\lambda_i = \mu_i$ for all i . □

(With Def. 8 this gives Stelling 4.2.5: w.r.t. a basis every vector of S has unique coordinates.)

Stelling 4.2.2 (Hoofdstelling) — n vectors span $\Rightarrow$ rank $\leq n$

If $S \subseteq \text{Span}(\mathbf{b}_1, \dots, \mathbf{b}_n)$ then $\text{rang}(S) \leq n$.

The lecturer: “bewijs is best technisch”, done at the end if time allows (see §6 below). The *idea* is the one thing to remember: **any $n + 1$ vectors in the span of n vectors are dependent**, and this comes down to a homogeneous system with more unknowns than equations (Stelling 3.5.2).

Consequence: every subset of $\mathbb{R}^n$ has rank $\leq n$, because $\mathbb{R}^n = \text{Span}(\mathbf{e}_1, \dots, \mathbf{e}_n)$.

Stelling 4.2.6 — every basis has the same size, namely the rank

Let $\text{rang}(S) = n$.

(1) **Every basis of S has exactly n elements.** In words: let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_m\}$ be a basis.
 - $\mathbf{b}_1, \dots, \mathbf{b}_m$ are independent, so $m \leq n$ (by the *definition* of rank as the maximum).
 - $S \subseteq \text{Span}(\mathbf{b}_1, \dots, \mathbf{b}_m)$, so by the Hoofdstelling $n = \text{rang}(S) \leq m$.
 - Hence $n = m$.

(2) **Every independent set of n vectors in S is a basis of S .** - Independent: given. - Does it span S ? Suppose not. Then there is an $\mathbf{s} \in S$ independent of the n vectors, giving $n + 1$ independent vectors in S , so $\text{rang}(S) \geq n + 1$: contradiction.

Stelling 4.2.8 — pivot columns form a basis

This is exactly the worked example in §3 (“hebben we dinsdag in ons voorbeeld gezien”): put the vectors as columns, reduce, the pivot columns (of the original matrix) form a basis, #pivots = rank.

Stelling 4.2.9 — bases exist and can be extended

For non-empty $S \subseteq S' \subseteq \mathbb{R}^n$: 1. If $S \neq \{\mathbf{0}\}$ then S has a basis. 2. Every basis of S can be extended to a basis of S' . 3. $\text{rang}(S) \leq \text{rang}(S')$.

Bewijs in woorden (1). Take $\mathbf{b}_1 \in S$ ($\neq \mathbf{0}$). If $\mathbf{b}_1$ spans S : done. If not, there is $\mathbf{b}_2 \in S$ independent of $\mathbf{b}_1$. If $\mathbf{b}_1, \mathbf{b}_2$ span S : done. If not, there is $\mathbf{b}_3 \in S$ independent of $\mathbf{b}_1, \mathbf{b}_2$, etc. This process **stops**, because $\text{rang}(S) \leq n$ (Hoofdstelling): you can never have more than n independent vectors. What you have when it stops is a basis. (2) is the same process, started from the given basis of S and picking new vectors from S' . (3) follows from (2).

Stelling 4.2.10 — rank does not change under these moves

In words: **if you add to S , or remove from S , a vector that is dependent on the (other) vectors of S , the rank does not change.** (The dictaat phrases it as: replacing $\mathbf{v}$ by $\lambda \mathbf{v}$, $\lambda \neq 0$, or by $\mathbf{v} + \lambda \mathbf{w}$, keeps the rank — these are exactly column versions of the row operations.)

6. Proof of the Hoofdstelling 4.2.2 (slide, “als er tijd is”)

Idea. Take $n + 1$ vectors $\mathbf{s}_1, \dots, \mathbf{s}_{n+1} \in S$ and show they are dependent by producing a non-trivial relation. Then no $n + 1$ vectors of S are independent, so $\text{rang}(S) \leq n$.

How to find the relation? Each $\mathbf{s}_i$ lies in $\text{Span}(\mathbf{b}_1, \dots, \mathbf{b}_n)$, so write it out:

$$\begin{aligned} \mathbf{s}_1 &= a_{11}\mathbf{b}_1 + a_{12}\mathbf{b}_2 + \dots + a_{1n}\mathbf{b}_n && \rightarrow \bar{\mathbf{a}}_1 = (a_{11}, \dots, a_{1n})^t \\ \mathbf{s}_2 &= a_{21}\mathbf{b}_1 + \dots + a_{2n}\mathbf{b}_n && \rightarrow \bar{\mathbf{a}}_2 \\ &\vdots && \\ \mathbf{s}_{n+1} &= a_{n+1,1}\mathbf{b}_1 + \dots + a_{n+1,n}\mathbf{b}_n && \rightarrow \bar{\mathbf{a}}_{n+1} \end{aligned}$$

We obtain $n + 1$ **coefficient vectors** $\bar{\mathbf{a}}_1, \dots, \bar{\mathbf{a}}_{n+1}$ in $\mathbb{R}^n$. Put them as columns of an $n \times (n + 1)$

matrix:

$$\begin{pmatrix} a_{11} & a_{21} & \cdots & a_{n+1,1} \\ \vdots & & & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{n+1,n} \end{pmatrix} = (\bar{\mathbf{a}}_1 \cdots \bar{\mathbf{a}}_{n+1}).$$

After Gauss–Jordan there are **at most** n **pivots** (only n rows), hence at most n independent columns among $n+1$. So there is a non-trivial relation between the columns:

$$\lambda_1 \bar{\mathbf{a}}_1 + \cdots + \lambda_{n+1} \bar{\mathbf{a}}_{n+1} = \mathbf{0}, \quad \text{not all } \lambda_i = 0.$$

Now compute, with the **same** λ_i :

$$\begin{aligned} \lambda_1 \mathbf{s}_1 + \cdots + \lambda_{n+1} \mathbf{s}_{n+1} &= \lambda_1 (a_{11} \mathbf{b}_1 + \cdots + a_{1n} \mathbf{b}_n) + \cdots \\ &\quad + \lambda_{n+1} (a_{n+1,1} \mathbf{b}_1 + \cdots + a_{n+1,n} \mathbf{b}_n) \\ &= (\lambda_1 a_{11} + \lambda_2 a_{21} + \cdots + \lambda_{n+1} a_{n+1,1}) \mathbf{b}_1 + \cdots \\ &\quad + (\lambda_1 a_{1n} + \cdots + \lambda_{n+1} a_{n+1,n}) \mathbf{b}_n \\ &= 0 \cdot \mathbf{b}_1 + \cdots + 0 \cdot \mathbf{b}_n = \mathbf{0}, \end{aligned}$$

because each bracket is one coordinate of $\lambda_1 \bar{\mathbf{a}}_1 + \cdots + \lambda_{n+1} \bar{\mathbf{a}}_{n+1} = \mathbf{0}$. So $\mathbf{s}_1, \dots, \mathbf{s}_{n+1}$ are dependent. Hence $\text{rang}(S) \leq n$. $\square$

Zelftest

1. Are $(1, 2)^t$ and $(2, 4)^t$ independent? *Answer:* no: $2(1, 2)^t - (2, 4)^t = \mathbf{0}$ is a non-trivial relation.

2. Are $(1, 0, 1)^t, (0, 1, 1)^t, (1, 1, 0)^t$ independent? *Answer:* columns into A , reduce: $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \sim$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{pmatrix}: \text{ 3 pivots, so only the trivial relation: independent, and they form a basis of } \mathbb{R}^3 \text{ (Stelling 4.2.6(2)).}$$

3. Can 4 vectors in $\mathbb{R}^3$ be independent? *Answer:* no — Hoofdstelling with $\mathbb{R}^3 = \text{Span}(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$, $\text{rank} \leq 3$. Concretely: A is 3×4 , at most 3 pivots, so a free variable gives a non-trivial relation.

4. $A' = \begin{pmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & 2 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ came from A with columns $\mathbf{v}_1, \dots, \mathbf{v}_4$. Write $\mathbf{v}_4$ in terms of the basis.

Answer: $\mathbf{v}_4 = -\mathbf{v}_1 + 6\mathbf{v}_2$ (read column 4 of A').

5. W is a subspace with basis of 3 vectors. Someone gives you 3 other independent vectors in W . Are they a basis of W ? *Answer:* yes, Stelling 4.2.6(2): $\dim W = 3$ and any 3 independent vectors in W form a basis.
6. State the difference between “ $\mathbf{w}$ is afhankelijk van $\mathbf{v}_1, \dots, \mathbf{v}_r$ ” and “ $\mathbf{v}_1, \dots, \mathbf{v}_r$ is afhankelijk”. *Answer:* the first is about one extra vector lying in the span (def. 3); the second is a property of the set: some member lies in the span of the others (def. 4), equivalently a non-trivial relation exists (Conclusie 2).